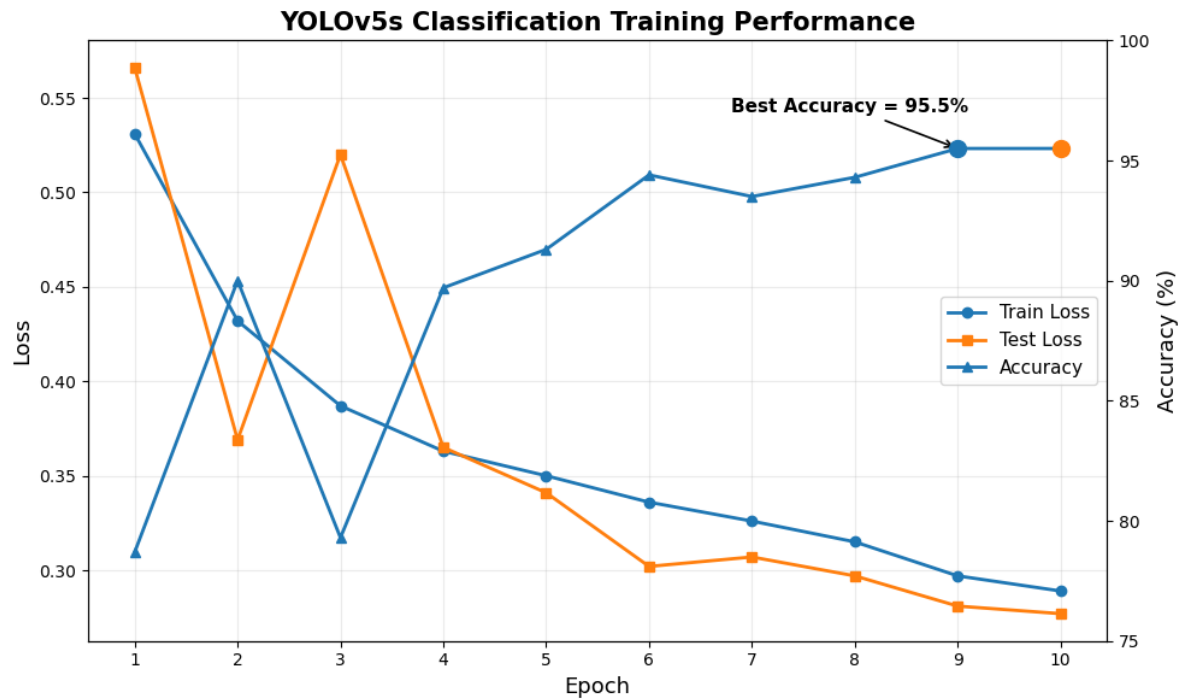


Code Details

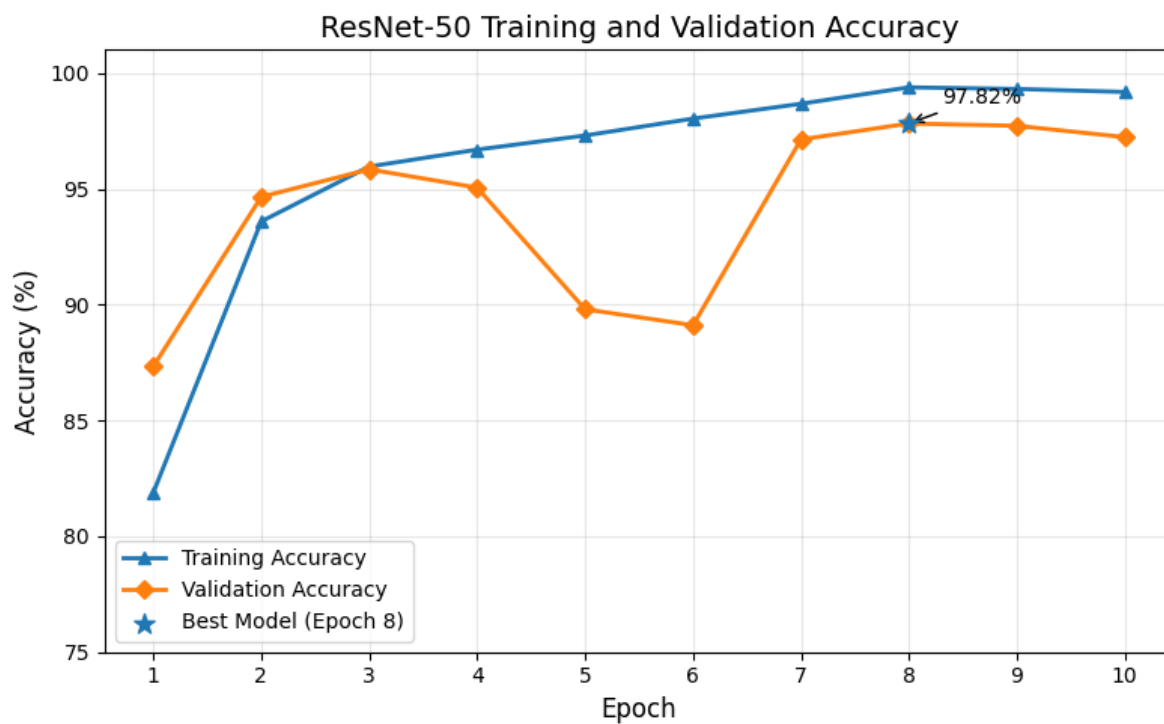
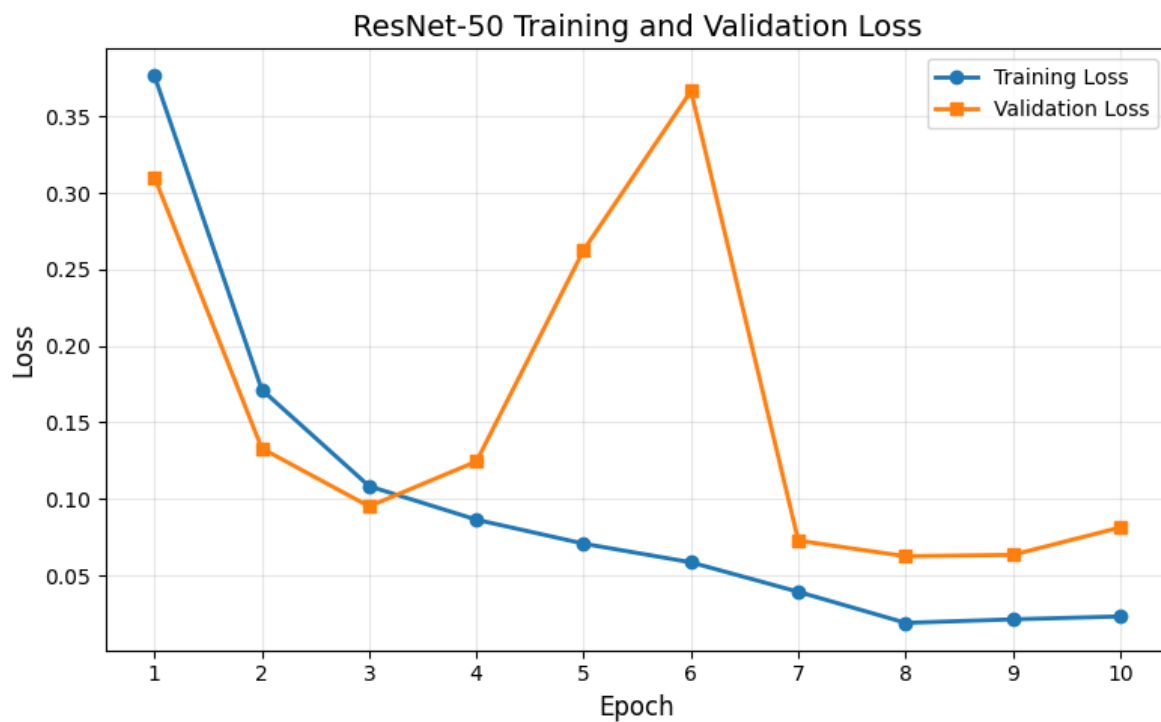
Dataset Celeb-DF-v2 Image Dataset
Total Real Images 5,036
Total Fake Images 5,067
Total Images 10,103

Parameter	Value	YOLO V5 hyperparameters
Training / Validation / Testing Split	70% / 10% / 20%	
Classification Classes	2 (Real, Fake)	
Model Architecture	YOLOv5s-cls	
Pretrained Weights	yolov5s-cls.pt	
Number of Epochs	10	
Batch Size	64	
Input Image Size	224 × 224 pixels	
Optimizer	Adam	
Initial Learning Rate (lr_0)	0.001	
Weight Decay ($decay$)	5×10^{-5}	
Label Smoothing	0.1	
Framework	PyTorch 2.11.0 + CUDA 12.8	
YOLOv5 Version	v7.0	
Model Parameters	4,175,042	



Parameter	
Input Image Size	224 × 224
Batch Size	32
Optimizer	Adam
Learning Rate	0.0001 (1×10^{-4})
Loss Function	Cross-Entropy Loss
Pretrained Weights	ResNet50_Weights.DEFAULT
Number of Classes	2
Random Split Seed	42
Training Epochs	10
Model Selection Criterion	Highest validation accuracy

Parameter	Value / Setting
Model Architecture	ResNet-50
Pretrained Model	Yes
Pretrained Weights	ImageNet pretrained ResNet-50
Number of Classes	2



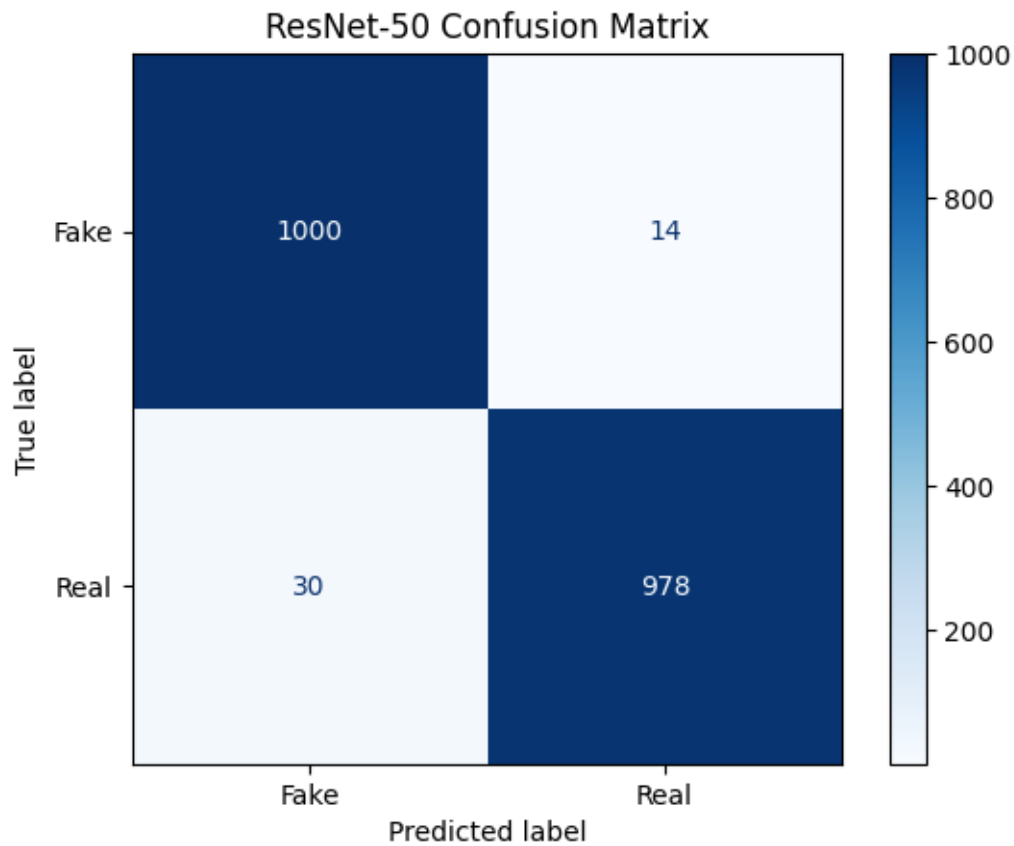
CLASSIFICATION REPORT

	precision	recall	f1-score	support
Fake	0.9709	0.9862	0.9785	1014
Real	0.9859	0.9702	0.9780	1008
accuracy			0.9782	2022
macro avg	0.9784	0.9782	0.9782	2022

weighted avg 0.9784 0.9782 0.9782 2022

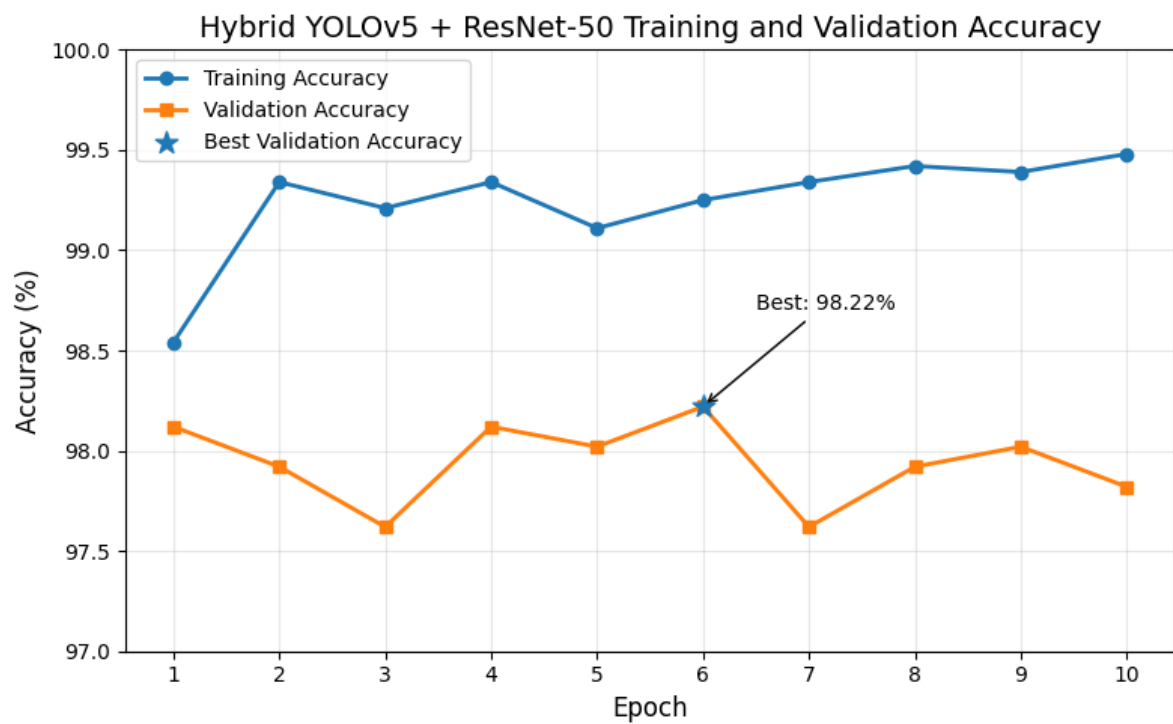
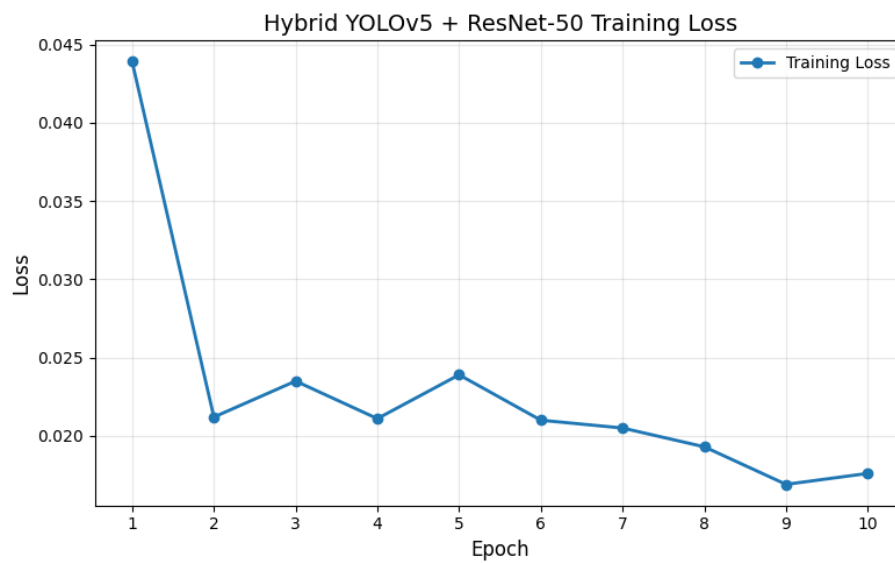
Confusion Matrix:

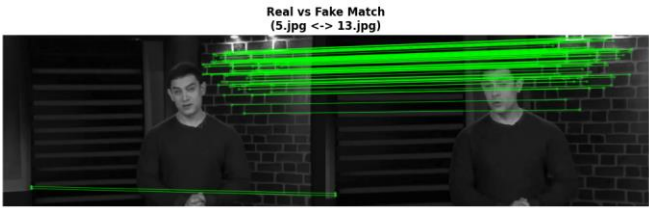
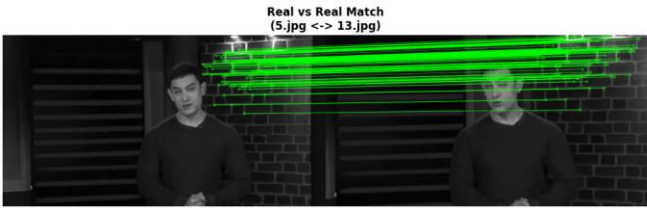
```
[[1000   14]
 [   30  978]]
```



Parameter	Value
Model Architecture	Hybrid YOLOv5 + ResNet-50
YOLOv5 Backbone	YOLOv5 Classification Model
ResNet Architecture	ResNet-50
YOLOv5 Feature Dimension	1280
ResNet-50 Feature Dimension	2048
Combined Feature Dimension	3328
Number of Classes	2 (Fake, Real)
ResNet-50 Training Epochs	5
Hybrid Model Training Epochs	10
YOLOv5 Parameters	4,169,250
YOLOv5 GFLOPs	10.4
YOLOv5 Pretrained Weights	best.pt
ResNet-50 Initialization	Pretrained ResNet-50
Computing Device	CUDA
GPU	NVIDIA Tesla T4
GPU Memory	14,913 MiB

Parameter	Value
PyTorch Version	2.11.0+cu128
Python Version	3.12.13
Fusion Strategy	Feature-level fusion
Classifier Input Dimension	3328
Hybrid Training Epochs	10





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EXTRACTED FEATURE MATCHING METRICS TABLE

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Comparison Type	Image 1 Name	Image 2 Name	Keypoints (Img 1)	Keypoints (Img 2)	Total Matches	Good Matches (Ratio Test)	Inlier Matches (RANSAC)	Inlier Ratio (%)	Homography Found
Real vs Real	5.jpg	13.jpg	357	265	357	145	104	71.72	Yes
Real vs Fake	5.jpg	13.jpg	357	262	357	127	99	77.95	Yes